MedChat - Healthcare RAG Assistant

AZIRO TECHNOLOGIES, PUNE

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202000801 - INDUSTRIAL INTERNSHIP

INTRODUCTION & PROBLEM STATEMENT

PROBLEM

- Patients struggle to understand medical reports & clinical terminology
- Standard chatbots hallucinate and lack domain grounding
- Keyword search misses relational context(e.g., Drug → Side Effect → Food)

SOLUTION

- Hybrid RAG + GraphRAG assistant grounded in peer-reviewed medical literature
- Structured knowledge graph (Neo4j) for relationship-aware reasoning
- Patient report upload → personalized, context-aware answers
- Knowledge base: Melanoma, Breast, Lung Cancer, Leukemia & Osteosarcoma

SYSTEM ARCHITECTURE

Medical PDFs → Doc Ingestion & Chunking → Chunks + Embeddings → Entity Extractor

Qdrant Vector DB
BM25 + Dense Embeddings
(all-MiniLM-L6-v2)
Hybrid Semantic Search

Neo4j Aura Knowledge Graph
99 Nodes | 153 Relationships
Cancer, Drug, Food, SideEffect, NutritionPattern, Symptom, Person

Context Builder (RAG + Graph Merge)

Groq LLM (LLaMA 3 / Mixtral)

TOOLS & TECHNOLOGIES

Language	Python 3.11.9
Vector DB	Qdrant (local persistent)
Graph DB	Neo4j Aura (cloud)
LLM	Groq API – LLaMA 3 / openai/gpt-oss-120b
Embeddings	sentence-transformers (MiniLM-L6-v2)
Hybrid Search	BM25 + Cosine Similarity
UI	Streamlit
PDF Parsing	Docling + PyMuPDF (fitz)

GRAPHRAG INTEGRATION

GraphRAG - Knowledge Graph Traversal

Breast Cancer
TREATED_WITH ▼
Paclitaxel
CAUSES ▼
Peripheral Neuropathy
WORSENS ▼
Fatigue
HELPS ▼
Protein Shake

METHODOLOGY & KEY FEATURES

RAG Pipeline

Medical PDFs ingested, chunked and stored in Qdrant. BM25 + dense embedding hybrid retrieval ensures high recall for clinical queries.

Entity Extractor

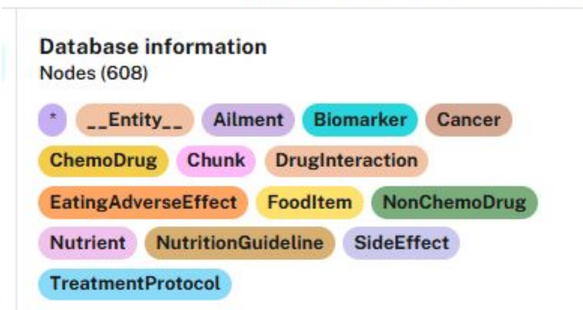
Regex + alias dictionaries (100+ medical aliases). Covers 6 cancers, 30+ drugs, 30+ side effects, 35+ foods. No external NLP APIs.

Patient Report Upload

Users upload personal reports (PDF). System extracts context, enriches query, and returns personalized, grounded responses.

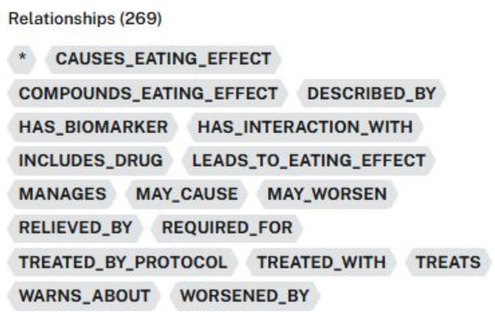
Multimodal Retrieval

Images from medical documents retrieved via PyMuPDF. Future: CLIP-based image embeddings for visual Q&A.



RESULTS & FEATURES

5	99	153	100+
Cancer Knowledge Domains	Graph Nodes	Graph Relationships	Medical Aliases



CHALLENGES & FUTURE SCOPE

Challenges Faced

- Balancing BM25 + embedding weights for optimal recall
- Manual curation of 100+ medical aliases without external NLP
- Graph schema design to cover multi-hop relationships

Future Scope

- RAGAS evaluation framework for retrieval accuracy metrics
- Multimodal RAG with CLIP-based image embeddings
- Conversational memory + Dockerized cloud deployment

CONCLUSION

A hybrid RAG + GraphRAG healthcare assistant that grounds clinical answers in peer-reviewed literature and multi-hop graph reasoning. It empowers patients to understand cancer diagnoses, treatments, drug interactions, and dietary guidance from uploaded medical reports—reducing hallucination risks of standard LLMs.

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